

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

FIRST COME FIRST SERVE SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>

void main()
{
int n,i,a[20],b[20],c[20],t=0,tt=0,wt=0;
float awt,att;
clrscr();
printf("\nFIRST COME FIRST SERVE SCHEDULING\n");
printf("\nEnter the Number of Processes : ");
scanf("%d",&n);
for(i=0;i<n;i++)
{
printf("\nEnter the Burst - Time of Process %d : ",i+1);
scanf("%d",&a[i]);
}
for(i=0;i<n;i++)
{
c[i]=t;
t=t+a[i];

b[i]=t;
}
```

```

printf("\n\nPROCESS\tBURST TIME\tWAITING TIME\tTURN AROUND TIME\n");
for(i=0;i<n;i++)
printf("\nP%d\t\t%d\t\t%d\t\t%d",i+1,a[i],c[i],b[i]);
for(i=0;i<n;i++)
tt=tt+b[i];
for(i=0;i<n;i++)
wt=wt+c[i];
awt=wt/n;
printf("\n\nThe Average Waiting Time : %f ms",awt);
att=tt/n;
printf("\n\nThe Average Turn Around Time : %f ms\n\n",att);
printf("\n\t\tGANTT CHART\n\n");
for(i=0;i<n;i++)
printf("\tP%d",i+1);
printf("\n-----\n");
printf("0");
for(i=0;i<n;i++)
printf("\t%d",b[i]);
printf("\n-----\n");
getch();
}

```

OUTPUT:

```
FIRST COME FIRST SERVE SCHEDULING

Enter the number of processes:3
Enter the burst time of process 1:24
Enter the burst time of process 2:3
Enter the burst time of process 3:3

Process Burst time      Waiting time      Turn around time

P1        24              0              24
P2         3             24              27
P3         3             27              30

The average waiting time:17.000000 ms
The average turn around time:27.000000 ms

                        GANTT CHART

      P1      P2      P3
-----
0         24      27      30
-----
```

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SHORTEST JOB FIRST SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int wt,j,c[10],p[10],i,tt,n,t1,t2,t3,a[10],ch;
    float awt=0.0,avg=0.0,s,t;
    clrscr();
    printf("\t\tSHORTEST JOB FIRST SCHEDULING\n\n");
    printf("Enter the Number of Processes : ");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        printf("\nEnter the Burst-Time for Process %d : ",i);
        scanf("%d",&c[i]);
        a[i]=c[i];
    }
    for(i=1;i<=n;i++)
    for(j=i+1;j<=n;j++)
    if(c[i]>c[j])
    {

        t1=c[i];
```

```

        c[i]=c[j];
        c[j]=t1;
    }
    i=1;j=1;c[0]=0;
    while(i<=n)
    {
        if(c[i]==a[j]&&ch!=j)
        {
            p[i]=j;
            ch=j;
            i++;
            j=0;
        }
        j++;
    }
    wt=0;c[0]=0;tt=0;
    printf("\n\nPROCESS\tBURST TIME\tWAITING TIME\tTURN AROUND TIME\n");
    for(i=0;i<=n-1;i++)
    {
        wt=wt+c[i];
        tt=tt+c[i+1];
        printf("\nP%d\t\t%d\t\t%d\t\t%d\n",p[i+1],c[i+1],wt,tt);
        awt=awt+wt;

    }
    avg=avg+tt;
    s=awt/n;

```

```

t=avg/n;
printf("\n\nThe Average Waiting Time : %f ms",s);
printf("\n\nThe Average Turn Around Time : %f ms",t);
printf("\n\n\n\t\tGANTT CHART\n");
printf("\n-----\n");
wt=0;
for(i=1;i<=n;i++)
    printf("\tP%d\t",p[i]);
printf("\n-----\n");
printf("0\t");
for(i=1;i<=n;i++)
{
    wt=wt+c[i];
    printf("%d\t",wt);
}
getch();
}

```

OUTPUT:

```
Shortest job first scheduling
Enter the number of processes:4
Enter the burst time for process1:6
Enter the burst time for process2:8
Enter the burst time for process3:7
Enter the burst time for process4:3_
Process          Brurst time      Waiting time      Turn around time
p4               3               0               3
p1               6               3               9
p3               7               9              16
p2               8              16              24

The average waiting time:7.000000 ms
The average turn around time:13.000000 ms
Gantt chart
-----
p4      p1      p3      p2
-----
0        3        9       16       24
```

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PRIORITY SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i,j,n,p[10],c[10],wt,tt,t1,t2,b[10],pro[10],tp,b1[10],ch,tb;
    float awt=0.0,s=0.0,avg=0.0,t=0.0;
    printf("\t\tPRIORITY SCHEDULING\n");
    printf("\nEnter the number of processes:");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        printf("\nPROCESS%d: \nPRIORITY:",i);
        scanf("%d",&p[i]);
        printf("\nBURST TIME:");
        scanf("%d",&c[i]);
        b[i]=p[i];
        b1[i]=p[i];
    }
    for(i=1;i<=n;i++)
    for(j=i+1;j<=n;j++)
    if(p[i]>p[j])
```



```

{
    t1=c[i];
    c[i]=c[j];
    c[j]=t1;
    t2=p[i];
    p[i]=p[j];
    p[j]=t2;
    tp=pro[i];
    pro[i]=pro[j];
    pro[j]=tp;
    tb=b[i];
    b[i]=b[j];
    b[j]=tb;
}
i=1;j=1;
while(i<=n)
{
    if(p[i]==b1[j]&&ch!=j)
    {
        p[i]=j;
        pro[i]=j;
        ch=j;
        i++;
        j=0;
    }
    j++;
}

```

```

wt=0;c[0]=0;tt=0;

printf("\n\nPROCESS\tBURST TIME\tWAITING TIME\tTURN AROUND
TIME\n");

for(i=0;i<n;i++)
{
    wt=wt+c[i];
    tt=tt+c[i+1];
    printf("\nP%d\t%d\t%d\t%d\t%d\n",pro[i+1],b[i+1],c[i+1],wt,tt);
    awt=awt+wt;
    avg=avg+tt;
}

s=awt/n;
t=avg/n;

printf("\n\nTHE AVERAGE WAITING TIME:%f ms",s);
printf("\n\nTHE AVERAGE TURN AROUND TIME:%f ms",t);
printf("\n\n\t\tGANTT CHART \n\n");

wt=0;
printf("\n-----\n");
printf("0");
for(i=0;i<=n;i++)
{
    wt=wt+c[i];
    printf("%d\t",wt);
}
}

```

OUTPUT:

```
PRIORITY SCHEDULING
Enter the number of processes:5

Process 1:
Priority:3
Burst time:10

Process 2:
Priority:1
Burst time:1

Process 3:
Priority:4
Burst time:2

Process 4:
Priority:5
Burst time:1

Process 5:
Priority:2
Burst time:5  _
```

Process	Priority	Burst time	Waiting time	Turn around time
P2	1	1	0	1
P5	2	5	1	6
P1	3	10	6	16
P3	4	2	16	18
P4	5	1	18	19

The average waiting time:8.200000 ms
The average turn around time:12.000000 ms

GANTT CHART

	P1	P2	P3	P4	P5
0	1	6	16	18	19

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ROUND ROBIN SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i,z=1,z1,n,n1,awt,p[10],j[10],t[10],t1[10],b[10],w[10],b1[10],b2[10];
    int temp[10],m=0,ts;
    float aw=0.0,at=0.0;
    printf("\n\t\t ROUND ROBIN SCHEDULING\n");
    printf("\n enter the number of processes:");
    scanf("%d",&n);
    n1=n;
    printf("\n enter the time slice:");
    scanf("%d",&ts);
    for(i=1;i<=n;i++) {
        printf("\n enter the burst time for p%d:",i);
        scanf("%d",&b[i]);
        b1[i]=b2[i]=b[i];p[i]=i;temp[i]=0;w[i]=0;t[i]=0;
    }
    printf("\n\n\t\t GANTT CHART\n");
    printf("\n-----\n");
    while(n1!=0)
    {
```

```

        for(i=1;i<=n;i++)
            if(b1[i]>ts)
            {
                m=m+ts;
                t1[z]=m;
                b1[i]=b1[i]-ts;
printf("\t p%d",p[i]);
                temp[i]++;
                z++;
            }
            else if(b1[i]!=0)
            {
                n1--;
                w[i]=m-temp[i]*ts;
printf("\t p%d",p[i]);
                aw+=w[i];m+=b1[i];t1[z]=m;t[i]=m;
                at=at+t[i];b1[i]=0;z++;
            }
        }

printf("\n-----\n");

        z1=z;

printf("\n0");

        for(i=1;i<z1;i++)
printf("\t %d",t1[i]);

printf("\n-----\n");

printf("\n\n PROCESS\t BURST TIME\t WAITING TIME\tTURN AROUND
TIME\n");

        for(i=1;i<=n;i++)

```

```
printf("p%d\tt%d\tt%d\tt%d\tt\n",p[i],b[i],w[i],t[i]);  
    at/=n;  
aw/=n;  
printf("\n the average waiting time:%fms",aw);  
printf("\n the average turn around time:%fms",at);  
getch();  
}
```

OUTPUT:

```
ROUND ROBIN SCHEDULING
Enter the number of processes:3

Enter the time slice:4

Enter the burst time for P1:24
Enter the burst time for P2:3
Enter the burst time for P3:3

          GANTT CHART

    P1      P2      P3      P1      P1      P1      P1      P1
-----
0         4         7         10        14        18        22        26        30
-----
Process      Burst time  Waiting time  Turn around time
P1            24           6             30
P2             3           4              7
P3             3           7             10
The average waiting time:5.666667 ms
The average turn around time:15.666667 ms
```

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PRODUCER CONSUMER PROBLEM

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

int product,counter=0,b,ch,buffer[100],in=0,out=0;

void producer();
void consumer();

void main()
{
printf("\nPRODUCER CONSUMER PROBLEM\n");
printf("\nEnter the Buffer size...:");
scanf("%d",&b);
while(ch!=3)
{
printf("\nMenu\n1.Product\n2.Consumer\n3.Exit\n");
printf("\nEnter your choice..:");
scanf("%d",&ch);
    switch(ch)
    {
        case 1:
producer();
            break;
        case 2:
```



```
consumer();
    break;
case 3:
exit(0);
    break;
default:
printf("wrong entry");
    }
}
}
void producer()
{
    while(counter==b)
    {
printf("\nBuffer Full...");
        return;
    }
printf("\nEnter Product Id..:");
scanf("%d",&product);
    buffer[in]=product;
    in=(in+1)%b;
    counter++;
}
void consumer()
{
    while(counter==0)
    {
```

```
printf("\nBuffer Empty...");  
    return;  
}  
    product=buffer[out];  
printf("\nConsumer Consumed %d Product",product);  
    out=(out+1)%b;  
    counter--;  
}
```

OUTPUT

```
PRODUCER CONSUMER PROBLEM

Enter the Buffer size...:2

Menu
1.Product
2.Consumer
3.Exit

Enter your choice...:1

Enter Product Id...:3

Menu
1.Product
2.Consumer
3.Exit

Enter your choice...:1

Enter Product Id...:4

Menu
1.Product
2.Consumer
3.Exit

Enter your choice...:1

Buffer Full....

Menu
1.Product
2.Consumer
3.Exit

Enter your choice...:2

Consumer Consumed 3 Product
Menu
1.Product
2.Consumer
3.Exit
```

Enter your choice..:2

Consumer Consumed 4 Product

Menu

1.Product

2.Consumer

3.Exit

Enter your choice..:2

Buffer Empty...

Menu

1.Product

2.Consumer

3.Exit

Enter your choice..:3

Process returned 0 (0x0) execution time : 25.650 s

Press any key to continue.

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READER WRITER PROBLEM

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int p[20],s[20],i,no,flag,rd[20],count;
    printf("\nREADER-WRITER PROBLEM\n");
    printf("\nEnter the process Number..");
    scanf("%d",&no);
    printf("-----");
    printf("\n\t0---READING");
    printf("\n\t1---WRITING");
    printf("\n-----");
    for(i=1;i<=no;i++)
    {
        printf("\nValue of process Id...");
        scanf("%d",&p[i]);
        printf("\nStatus of Process Id...");
        scanf("%d",&s[i]);
    }
    printf("\nProcess Id\tStatus");
    for(i=1;i<=no;i++)
    {
```

```

printf("\n%d\t\t%d",p[i],s[i]);
    }
printf("\n\nWriters list");
    flag=0;
    for(i=1;i<=no;i++)
    {
        if(s[i]==1)
        {
            flag=1;
printf("\nProcess Id %d",p[i]);
            count++;
            while(count!=0)
            {
                count--;
            }
        }
        else
        {
rd[i]=p[i];
        }
    }
    if(flag==1)
printf("\nWriting is completed");
    else
printf("\nno Writing Process");
printf("\n\nReaders List");
    flag=0;

```

```
for(i=1;i<=no;i++)
{
    if(s[i]!=1)
    {
        flag=1;
printf("\nProcess Id=%d",rd[i]);
    }
}
if(flag==0)
{
printf("\nNo Reading Process");
}
}
```

OUTPUT:

READER-WRITER PROBLEM

Enter the process Number..3

0---READING

1---WRITING

Value of process Id...8

Status of Process Id...1

Value of process Id...5

Status of Process Id...0

Value of process Id...6

Status of Process Id...0

Process Id	Status
------------	--------

8	1
---	---

5	0
---	---

6	0
---	---

Writers list

Process Id 8

Writing is completed

Readers List

Process Id=5

Process Id=6

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DINING PHILOSOPHER PROBLEM

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
int st[25],temp[25];
void eating(int,int,int);
void main()
{
    int n=5,f,i,x,y;
    int ch;
    printf("\n\t\tDINING-PHILOSPER PROBLEM");
    xx:
    printf("\n\nASSUMPTION\n");
    printf("\nHUNGRY==>1\nEATING==>2\nTHINKING==>3\n");
    for(i=1;i<=n;i++)
    do {
        f=0;
        printf("\nEnter the status of %d :",i);
        scanf("%d",&st[i]);
        if(st[i]>3||(st[i]<1))
        {
            printf("\nEntered philosopher ststus is invalid");
            printf("\nDo you want to try again(Y/N):");
            ch=getch();
```

```

        if(ch=='y'||ch=='Y')
            f=1;
        else
        {
            f=0;
printf("\nSTOP");
getch();
exit(1);
        }
    }
}
while(f==1);
for(i=1;i<=n;i++)
{
    if(i==n)
    {
        x=n;
        y=1;
    }
    else
    {
        x=i;
        y=i+1;
    }
    if((st[x]==2)&&(st[y]==2))
    {
printf("\nEntered philosopher stsus id invalid");

```

```

printf("\nSorry try again...");
goto xx;
    }
}
printf("\n\nRESULTS.....");
    for(i=1;i<=n;i++)
    {
        if(st[i]==3)
printf("\nPhilosopher %d is thinking\n",i);
        if(st[i]==2)
printf("\nPhilosopher %d id eating\n",i);
        if(st[i]==1)
        {
            if(i==1)
                eating(i+1,i+4,i);
            else eating(i+1,i-1,i);
        }
    }
}

void eating(int x,inty,int z)
{
    if(temp[x]==1)
    {
printf("\nPhilosopher %d is hungry and cannot eat since %d has the possibility of
eating\n",x,z);

        return;
    }
    if(temp[y]==1)

```

```

    {
printf("\nPhilosopher %d is hungry and cannot eat since %d has the possibility of
eating\n",z,y);

    return;

    }
    if((st[x]!=2)&&(st[y]!=2))
    {
printf("\n\nPhilosopher %d is hungry and cannot eat\n",z);
st[z]=2;

    temp[z]=1;

    }
    if((st[x]==2)&&(st[y]==2))
    {
printf("\n\nPhilosopher %d is hungry but not eatting since neighbours %d, %d are
eating\n",z,x,y);

    return;

    }
    if(st[x]==2)
    {
printf("\n\nPhilosopher %d is hungry but not eating since neighbours %d, \n",z,x);

    return;

    }
    if(st[y]==2)
    {
printf("\n\nPhilosopher %d is hungry but not eating since neighbours %d, \n",z,y);

    return;

    }
    }

```


OUTPUT:

```

                                DINING-PHILOSPER PROBLEM
ASSUMPTION
HUNGRY==>1
EATING==>2
THINKING==>3

Enter the status of 1 :1
Enter the status of 2 :2
Enter the status of 3 :3
Enter the status of 4 :1
Enter the status of 5 :2

RESULTS....:
Philosopher 1 is hungry but not eatting since neighbours 2, 5 are eating
Philosopher 2 id eating
Philosopher 3 is thinking

Philosopher 4 is hungry but not eating since neighbours 5,
Philosopher 5 id eating
```

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CONTIGUOUS MEMORY ALLOCATION

SOURCE CODE:

```
#include<stdio.h>
#include<stdlib.h>

int i,n;
struct f
{
    int nm,r;
}
h[25];
int fit()
{
    printf("\n enter the total no.of blocks and holes:");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        xx:
        printf("\n enter whether it is block or holes:");
        scanf("%d",&h[i].nm);
        if((h[i].nm!=1)&&(h[i].nm!=2))
        {
            printf("\n enter the name is invalid try again");
            goto xx;
        }
    }
}
```

```

printf("\n enter the size:");
scanf("%d",&h[i].r);
    }
    return 0;
}

int ffit(int x)
{
    for(i=0;i<=n;i++)
    {
        if((h[i].nm==2)&&(x<=h[i].r))
        {
            printf("\n allocation of block %d by first fit:",x);
            printf("%d",h[i].r);
                break;
            }
        }
        if(i==n+1)
        printf("\n first fit:");
        return 0;
    }

int wfit(int x)
{
    int t=0;
    for(i=0;i<=n;i++)
    {
        if((h[i].nm==2)&&(x<=h[i].r)&&(t<h[i].r))
            t=h[i].r;
    }
}

```



```

    }
    printf("\n allocation of block %d by worst fit:",x);
    printf("%d",t);
    return 0;
}
int bfit(int x)
{
    int temp[25],k=1;
    for(i=1;i<=n;i++)
    {
        if((h[i].nm==2)&&(x<=h[i].r))
        {
            temp[k]=h[i].r;
            k++;
        }
    }
    for(i=1;i<k;i++)
    {
        int j;
        for(j=i+1;j<k;j++)
        {
            if(temp[i]>temp[j])
            {
                int t=temp[i];
                temp[i]=temp[j];
                temp[j]=t;
            }
        }
    }
}

```

```

    }
}
printf("\n allocation of bloack %d by best fit:",x);
printf("%d",temp[1]);
return 0;
}
int show()
{
printf("\n allocation of block and holes \n");
for(i=1;i<=n;i++)
{
    if(h[i].nm==1)
    {
printf("block(%d)\n",h[i].r);
    }
    else
    {
printf("hole(%d)\n",h[i].r);
    }
}
return 0;
}
int main()
{
    int b;
printf("\n block =1\n");
printf("\n hole=2\n");

```

```
fit();  
show();  
printf("\n enter the block to be inserted:");  
scanf("%d",&b);  
ffit(b);  
    bfit(b);  
wfit(b);  
    return 0;  
}
```

OUTPUT:

```
block =1
hole=2
enter the total no.of blocks and holes:4
enter whether it is block or holes:1
enter the size:41
enter whether it is block or holes:1
enter the size:34
enter whether it is block or holes:2
enter the size:51
enter whether it is block or holes:2
enter the size:31

allocation of block and holes
block(41)
block(34)
hole(51)
hole(31)

enter the block to be inserted:22

allocation of block 22 by first fit:51
allocation of bloack 22 by best fit:31
allocation of block 22 by worst fit:51
Process returned 0 (0x0)   execution time : 24.929 s
Press any key to continue.
```

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SIMULATION OF PAGING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int phyadr,psi,frame,page,offset,n,pt[20],logadr,i;
clrscr();
printf("\nSIMULATION OF PAGING");
printf("\n\nEnter the Number of Pages...: ");
scanf("%d",&n);
printf("\n\nEnter the Page Size..: ");
scanf("%d",&psi);
printf("\n\nEnter the Page Table Values...");
printf("\n\nPage Number\tValue");
for(i=0;i<n;i++)
{
printf("\n%d\t\t",i);
scanf("%d",&pt[i]); }
printf("\n\nEnter the Logical Address..: ");
scanf("%d",&logadr);
if(logadr>(n*psi)-1)
printf("\nInvalid address...");
else {
```

```
offset=logadr%psi;
page=logadr/psi;
phyadr=(psi*pt[page]+offset);
printf("\n\nPhysical Address = (%d * %d) + %d",psi,pt[page],offset);
printf("\n\nPhysical Address is %d",phyadr);
}
getch();
}
```

OUTPUT:

```
SIMULATION OF PAGING
Enter the Number of Pages...: 5

Enter the Page Size..: 5

Enter the Page Table Values...
Page Number      Value
0                4
1                6
2                1
3                7
4                3

Enter the Logical Address..: 3

Physical Address = (5 * 7) + 3
Physical Address is 38|
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

SIMULATION OF SEGMENTATION

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>

int main()
{
    int n,x,y,sc,pa,ba,ab,li,b[10],l[10],i[10],j;
    y=y-1;
    printf("\n\tSIMULATION OF SEGMENTATION\n");
    printf("\nEnter the Number of Entries : ");
    scanf("%d",&n);
    for(x=0;x<n;x++)
    {
        printf("SEGMENT NO:");
        scanf("%d",&i[x]);
        printf("LIMIT:");
        scanf("%d",&l[x]);
        printf("BASE:");
        scanf("%d",&b[x]);
    }
    printf("\nEnter the Segment No.: ");
    scanf("%d",&sc);
    for(j=0;j<n;j++)
    if(i[j]==sc)
```



```
{
y=1;
ba=b[j];
li=l[j];
}
if(y<0)
{
printf("\nINVALID SEGMENT\n");
}
printf("\nEnter the Byte to be Referenced : ");
scanf("%d",&ab);
if(ab>li)
{
printf("\nCannot Find this Address :");
printf("\nLimit Only %d",i);
}
pa=ba+ab;
printf("\nThe Physical Address is : ");
printf("%d\n",pa);
getch( );
}
```

OUTPUT:

SIMULATION OF SEGMENTATION

Enter the Number of Entries : 4

SEGMENT NO:1

LIMIT:100

BASE:1200

SEGMENT NO:2

LIMIT:470

BASE:3600

SEGMENT NO:3

LIMIT:720

BASE:1000

SEGMENT NO:4

LIMIT:280

BASE:4500

Enter the Segment No.: 3

Enter the Byte to be Referenced : 700

The Physical Address is : 1700

|

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

FIFO PAGE REPLACEMENT

SOURCE CODE:

```
#include<stdio.h>
#include<stdlib.h>

void main()
{
int i,j,a[50],frame[10],n,no,k,avail,count=0;
printf("\n***FIFO PAGE REPLACEMENT ***");
printf("\n Enter the number of reference string:\n");
scanf("%d",&n);
printf("Enter the page number:\n");
for(i=1;i<=n;i++)
scanf("%d",&a[i]);
printf("\nEnter the number of frames:");
scanf("%d",&no);
for(i=0;i<=no;i++)
frame[i]=-1;
j=0;
printf("\n Ref string\tpage frame\n");
for(i=1;i<=n;i++)
{
printf("%d\t\t",a[i]);
avail=0;
for(k=0;k<=no;k++)
```

```
    if(frame[k]==a[i])
        avail=1;
    if(avail==0)
    {
        frame[j]=a[i];
        j=(j+1)%no;
        count++;
        for(k=0;k<no;k++)
            printf("%d\t",frame[k]);
    }
    printf("\n");
}
printf("\n page fault is %d",count);
getch ();
}
```

OUTPUT:

```
***FIFO PAGE REPLACEMENT ***
Enter the number of reference string:
20
Enter the page number:
7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1
Enter the number of frames:3

Ref string      page frame
7               7       -1      -1
0               7       0       -1
1               7       0       1
2               2       0       1
0
3               2       3       1
0               2       3       0
4               4       3       0
2               4       2       0
3               4       2       3
0               0       2       3
3
2
1               0       1       3
2               0       1       2
0
1
7               7       1       2
0               7       0       2
1               7       0       1

page fault is 15
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

LRU PAGE REPLACEMENT

SOURCE CODE:

```
#include<stdio.h>

void main()
{
int q[20],p[50],c=0,c1,f,i,j,k=0,n,r,t,b[20],c2[20];
printf("\n\t\t***LRU-PAGE REPLACEMENT***\n");
printf("\nEnter the no. of pages:");
scanf("%d",&n);
printf("\nEnter the reference string:");
for(i=0;i<n;i++)
    scanf("%d",&p[i]);
printf("\nEnter the no. of frames:");
scanf("%d",&f);
printf("\nReference string\t \tpage frames");
q[k]=p[k];
printf("\n%d",p[k]);
printf("\t%d",q[k]);
c++;
k++;
for(i=1;i<n;i++)
{
printf("\n%d",p[i]);
c1=0;
```

```
for(j=0;j<f;j++)
{
    if(p[i]!=q[j]) c1++;
}
if(c1==f)
{
    c++;
if(k<f)
{
    q[k]=p[i];
k++;
for(j=0;j<k;j++)
    printf("\t%d",q[j]);
}
else
{
    for(r=0;r<f;r++)
    {
        c2[r]=0;
for(j=i-1;j<n;j--)
{
    if(q[r]!=p[j])
        c2[r]++;
else
    break;
}
}
```

```

for(r=0;r<f;r++)
    b[r]=c2[r];
for(r=0;r<f;r++)
    b[r]=c2[r];
for(r=0;r<f;r++)
{
for(j=r;j<f;j++)
{
    if(b[r]<b[j])
    {
        t=b[r];
b[r]=b[j];
b[j]=t;
    }
}
}
for(r=0;r<f;r++)    {
    if(c2[r]==b[0])
        q[r]=p[i];
printf("\t%d",q[r]);
}
}
}
}

printf("\n\nThe no. of page faults is %d",c);
getch ();
}

```


OUTPUT:

```
***LRU-PAGE REPLACEMENT***

Enter the no. of pages:20

Enter the reference string:7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Enter the no. of frames:3

Reference string      page frames
7                     7
0
1                     7      1
2                     7      1      2
0                     0      1      2
3                     0      3      2
0
4                     0      3      4
2                     0      2      4
3                     3      2      4
0                     3      2      0
3
2
1                     3      2      1
2
0                     0      2      1
1
7                     0      7      1
0
1
The no. of page faults is 12
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

OPTIMAL PAGE REPLACEMENT

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#define max 25
void main()
{
int frame[10],length[10],index,highest;
int i,j,k,n,nf,page[max],temp;
int flag=0,pf=0,found=0;
printf("\n\toptimal PAGE REPLACEMENT\n");
printf("\nEnter no. of frames:");
scanf("%d",&nf);
for(i=0;i<nf;i++)
    frame[i]=-1;
printf("\nEnter the no. of ref string:");
scanf("%d",&n);
printf("\nEnter the ref string:");
for(i=0;i<n;i++)
{
scanf("%d",&page[i]);
}
printf("\n\tRef.String\tPage\t Frames\n");
for(i=0;i<n;i++)
{
```

```

        flag=0;
for(j=0;j<nf;j++)
{
if(frame[j]==page[i])
{
printf("\n%d\t\t",page[i]);
flag=1;
break;
}
}
if(flag==0)
{
for(j=0;j<nf;j++)
{
for(temp=i+1;temp<n;temp++)
{
length[j]=0;
if(frame[j]==page[temp])
{
length[j]=temp-i;
break;
}
}
} found=0;
for(j=0;j<nf;j++)
{
if(length[j]==0)

```

```

        {
            index=j;
found=1;
break;
        }
    }
    if(found==0)
    {
        highest=length[0];
        index=0;
        for(j=1;j<nf;j++)
        {
if(highest<length[j])
        {
            highest=length[j];
            index=j;
        }
    }
    frame[index]=page[i];
    printf("\n%d\t\t",page[i]);
    pf++;
    for(k=0;k<nf;k++)
        printf("%d\t",frame[k]);
    }
}
printf("\nNumber of page fault is:%d\n\n",pf);

```

```
getch ();
```

```
}
```

OUTPUT:

```
OPTIMAL PAGE REPLACEMENT
Enter no. of frames:3
Enter the no. of ref string:20
Enter the ref string:7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1
    Ref.String    Page    Frames
7                7        -1      -1
0                7         0      -1
1                7         0       1
2                2         0       1
0
3                2         0       3
0
4                2         4       3
2
3
0                2         0       3
3
2
1                2         0       1
2
0
1
7                7         0       1
0
1
Number of page fault is:9
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

CONTIGUOUS FILE ALLOCATION

SOURCE CODE:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
int i,j,b[20],sb[20],t[20],n,c[20][20],x;
clrscr();
printf("\t\t***CONTIGUOUS FILE ALLOCATION***\n");
printf("Enter the number of files:");
scanf("%d",&n);
for(i=0;i<n;i++)
{
printf("\nEnter the number of blocks occupied by %d:",i+1);
scanf("%d",&b[i]);
printf("\nEnter the starting block of file %d:",i+1);
scanf("%d",&sb[i]);
t[i]=sb[i];
for(j=0;j<b[i];j++)
c[i][j]=sb[i]++;
}
printf("\nFile name\tStart\tBlock Length\n");
for(i=0;i<n;i++)
printf("%d\t%d\t%d\t%d\n",i+1,t[i],b[i]);
printf("Enter the file name:");
```



```
scanf("%d",&x);  
printf("\nFile name is %d",x);  
printf("\nLength is %d",b[x-1]);  
printf("\nBlocks Occupied..");  
for(i=0;i<b[x-1];i++)  
printf("\t%d",c[x-1][i]);  
getch();  
return 0;  
}
```

OUTPUT:

```
***CONTIGUOUS FILE ALLOCATION***
Enter the number of files:5
Enter the number of blocks occupied by 1:2
Enter the starting block of file 1:0
Enter the number of blocks occupied by 2:3
Enter the starting block of file 2:14
Enter the number of blocks occupied by 3:6
Enter the starting block of file 3:19
Enter the number of blocks occupied by 4:4
Enter the starting block of file 4:28
Enter the number of blocks occupied by 5:2
Enter the starting block of file 5:6
File name      Start   Block Length
1              0       2
2              14      3
3              19      6
4              28      4
5              6       2
Enter the file name:1
File name is  1
Length is  2
Blocks Occupied..      0      1
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

LINKED FILE ALLOCATION

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
struct fileno
{
char fname[10];
int start,size,block[10];
}
f[10];
int main()
{
int i,j,n;
printf("\n\t\tLINKED FILE ALLOCATION");
printf("\nEnter the no. of files:\n");
scanf("%d",&n);
for(i=0;i<n;i++)
{
printf("\nEnter the file name:");
scanf("%s",&f[i].fname);
printf("\nEnter the starting block:");
scanf("%d",&f[i].start);

f[i].block[0]=f[i].start;
printf("\nEnter the no. of blocks:");
```

```
scanf("%d",&f[i].size);
printf("\nEnter the block numbers:");
for(j=1;j<=f[i].size;j++)
{
scanf("%d",&f[i].block[j]);
}
}

printf("\nFile\tStart\tSize\tBlock\n");
for(i=0;i<n;i++)
{
printf("%s\t%d\t%d\t",f[i].fname,f[i].start,f[i].size);
for(j=0;j<=f[i].size-1;j++)
printf("%d-->",f[i].block[j]);
printf("%d(-1)",f[i].block[j]);
printf("\n");
}

getch();
return 0;
}
```

OUTPUT:

```
LINKED FILE ALLOCATION
Enter the no. of files:
1
Enter the file name:jeep
Enter the starting block:9
Enter the no. of blocks:4
Enter the block numbers:16
1
10
25
File    Start   Size   Block
jeep    9        4      9-->16-->1-->10-->25(-1)
|
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

INDEXED FILE ALLOCATION

SOURCE CODE:

```
#include<stdio.h>
#include<stdlib.h>

int main()
{
    int n,m[20],i,j,sb[20],b[20][20],x;
    printf("\t\t***INDEXED FILE ALLOCATION***");
    printf("\nEnter no. of files:");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("\nEnter index %d:",i+1);
        scanf("%d",&sb[i]);
        printf("\nEnter no. of blocks occupied by file %d:",i+1);
        scanf("%d",&m[i]);
        printf("\nEnter blocks of file %d:",i+1);
        for(j=0;j<m[i];j++)
            scanf("%d",&b[i][j]);
    }
    printf("\nFile\tIndex\tLength\n");
    for(i=0;i<n;i++)
    {
        printf("%d\t%d\t%d\n",i+1,sb[i],m[i]);
    }
}
```

```
}  
printf("\nEnter file name:");  
scanf("%d",&x);  
printf("\nFile name is:%d",x);  
i=x-1;  
printf("\nIndex is:%d",sb[i]);  
printf("\nBlocks occupied are:");  
for(j=0;j<m[i];j++)  
printf("\t%d",b[i][j]);  
getch();  
return 0;  
}
```

OUTPUT:

```
***INDEXED FILE ALLOCATION***
Enter no. of files:1
Enter index 1:19
Enter no. of blocks occupied by file 1:5
Enter blocks of file 1:9
16
1
10
25
File      Index  Length
1         19     5
Enter file name:1
File name is:1
Index is:19
Blocks occupied are:  9      16      1      10      25|
```


REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

FIRST COME FIRST SERVE SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
#include<process.h>
int h,a[20],b=0,i,j,n;
void main( ) {
    printf("FIRST COME FIRST SERVE DISK SCHEDULING\n");
    printf("\nEnter the Number of Cylinder : ");
    scanf("%d",&n);
    printf("\nEnter the Cylinder Values : \n");
    for(i=1;i<=n;i++)
        scanf("%d",&a[i]);
    printf("\nEnter the Starting Head Position : ");
    scanf("%d",&h);
    printf("\nThe Order of Visiting the Cylinders:\n");
    for(i=1;i<=n;i++)
        printf("\t%d",a[i]);
    for(i=1;i<=n;i++) {
        b+=abs(h-a[i]);
        h=a[i];
    }
    printf("\nThe Seek Time is %d Cylinders",b);
    getch( );
}
```

OUTPUT:

```
*** FIRST COME FIRST SERVE DISK SCHEDULING***  
Enter the number of cylinder:8  
Enter the cylinder values:  
98  
183  
37  
122  
14  
124  
65  
67  
  
Enter the starting head position:53  
  
The order of visiting the cylinders:  
    98    183    37    122    14    124    65    67  
The seek time is 640 cylinders_
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

SHORTEST SEEK TIME FIRST SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
int sstf();
int seek=0,t1,t2,n,a[20]={ {0} },z=1,d[20]={ {0} },h=0;
void main()
{
    int i,j,hr;
    clrscr();
    printf("\t\t SHORTEST SEEK TIME FIRST SCHEDULING \n\n");
    printf("Enter the Number of Cylinders : ");
    scanf("%d",&n);
    printf("\nEnter the Cylinder Values : \n");
    for(i=1;i<=n;i++)
        scanf("%d",&a[i]);
    printf("\nEnter the Starting Head Position : ");
    scanf("%d",&h);
    hr=h;
    printf("\nThe Order of Visiting the Cylinders : \n");
    for(i=1;i<=n;i++)
        sstf();
    printf("\n");
    for(i=1;i<=n;i++)
    {
```

```

        seek=seek+abs(hr-d[i]);
        hr=d[i];
    }
    printf("\n\nThe Seek Time is %d",seek);
    getch();
}

int sstf()
{
    int b[20]={ {0} },c=1,i,j;
    for(i=1;i<=n;i++)
        b[i]=abs(h-a[i]);
    for(i=1;i<=n;i++)
        if(b[c]>b[i])
            c=i;
    printf("\t%d",a[c]);
    h=a[c];
    d[z]=a[c];
    z++;
    a[c]=9999;
    return(0);
}

```

OUTPUT:

```
          SHORTEST SEEK TIME FIRST SCHEDULING
Enter the number of cylinders:8
Enter the cylinder values:98 183 37 122 14 124 65 67
Enter the starting head position:53
The order of visiting the cylinders:
      65      67      37      14      98      122      124      183
The seek time is 236
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

SCAN SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
int hr,right=0,i,j,h,end,n,a[20],temp,post,left=0,start=0;
printf("\t\t**SCAN SCHEDULING**\n");
printf("\nEnter the ending position of disk:\n");
scanf("%d",&end);
printf("\nEnter the number of cylinders:\n");
scanf("%d",&n);
printf("\nEnter the cylinder value between 0 and %d...\n",end);
for(i=1;i<=n;i++)
{
scanf("%d",&a[i]);
}
for(i=1;i<=n;i++)
{
for(j=i+1;j<=n;j++)
{
if(a[i]>a[j])
{
temp=a[i];
a[i]=a[j];
a[j]=temp;
}
```

```

a[j]=temp;
}
}
}
printf("\nEnter the head position:\n");
scanf("%d",&h);
hr=h;
for(i=1;i<=n;i++)
{
if(h<a[i])
break;
}
post=i;
printf("\nMoving in the right side:\n");
printf("\nThe order of visiting the cylinders:\n");
for(i=post;i<=n;i++)
{
right=right+abs(h-a[i]);
h=a[i];
printf("\t%d",h);
}
right=right+abs(h-end);
h=end;
for(i=post-1;i>=1;i--)
{
printf("\t%d",h);
right=right+abs(h-a[i]);

```

```
h=a[i];
}
printf("\t%d",h);
printf("\n\nThe seek time is:%d\n",right);
printf("\n\nMoving in the left side:\n");
printf("\n\nThe order of visiting the cylinder:\n");
for(i=post-1;i>=1;i--)
{
left=left+abs(hr-a[i]);
hr=a[i];
printf("\t%d",hr);
}
left=left+abs(hr-start);
hr=start;
for(i=post;i<=n;i++)
{
printf("\t%d",hr);
left=left+abs(hr-a[i]);
hr=a[i];
}
printf("\t%d",hr);
printf("\n\nThe seek time is:%d\n",left);
getch();
return 0;
}
```


OUTPUT:

```

**SCAN SCHEDULING**

Enter the ending position of disk:
199

Enter the number of cylinders:
8

Enter the cylinder value between 0 and 199...
98
183
37
122
14
124
65
67

Enter the head position:
53

Moving in the right side:

The order of visiting the cylinders:
    65    67    98    122    124    183    199    37    14
The seek time is:331

Moving in the left side:

The order of visiting the cylinder:
    37    14    0    65    67    98    122    124    183
The seek time is:236

Process returned 0 (0x0)  execution time : 54.971 s
Press any key to continue.
```

REG NO: 21TD0005 NAME: AKASH L YEAR/SEM/SEC: Y3/S5/B

C-SCAN SCHEDULING

SOURCE CODE:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main( )
{
    int left=0,hr,right=0,start=0,i,j,h,end,n,a[20],b[20],temp,post;
    clrscr( );
    printf(" C – SCAN SCHEDULING ");
    printf("\n\nEnter the Ending Position of the Disc : ");
    scanf("%d",&end);
    printf("\nEnter the Number of Cylinders : ");
    scanf("%d",&n);
    printf("\nEnter the Cylinder Values between 0 and %d : \n",end);
    for(i=1;i<=n;i++)
        scanf("%d",&a[i]);
    for(i=1;i<=n;i++)
        for(j=i+1;j<=n;j++)
            if(a[i]>a[j])
            {
                temp=a[i];
                a[i]=a[j];
                a[j]=temp;
            }
    printf("\nEnter the Starting Head Position : ");
```

```

scanf("%d",&h);
hr=h;
for(i=1;i<=n;i++)
{
    if(h<a[i])
        break;
}
post=i;
printf("The Order of Visiting the Cylinders : \n");
for(i=post;i<=n;i++)
{
    right=right+abs(h-a[i]);
    h=a[i];
    printf("\t%d",h);
}
right=right+abs(h-end);
printf("\t%d",end);
right=right+abs(end-start);
h=start;
for(i=1;i<post;i++)
{
    printf("\t%d",h);
    right=right+abs(h-a[i]);
    h=a[i];
}
printf("\t%d",h);
printf("\n\nThe Seek Time is : %d\n\n",right);
getch( );
}

```

OUTPUT:

```
C-SCAN SCHEDULING
Enter the ending position of the disk:199
Enter the number of cylinders:8
Enter the cylinder values between 0 and 199:
98
183
37
122
14
124
65
67

Enter the head position:
53

The order of visiting the cylinders:
53 65 67 98 122 124 183 199 0 14 37

The seek time is:382
```